

**Macroeconomic Aggregates vs. Government Policy: An ARDL Cointegration Analysis
of U.S. Disposable Income Drivers (1951–2025)**

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Abstract

In this study, we perform a full-scale econometric analysis using a quarterly ARDL bounds testing framework to identify the underlying dynamics influencing U.S. disposable income from 1951 to 2025. Disposable income, as a financial measure of household economic conditions and national stability, is one of the key indicators of economic well-being. We assess the possible impacts of GDP, unemployment, personal consumption expenditures, and federal government transfers over the short and long run using quarterly time-series data ($n = 297$). After identifying mixed $I(0)$ and $I(1)$ variables with no $I(2)$ series, the AIC selected an $ARDL(3,3,0,3,4)$ model. The bounds test produced an F-statistic of 6.40, which exceeds the 5% $I(1)$ upper critical value of 4.85, indicating a long-run relationship among the variables. Under HAC-robust inference, GDP remains the only statistically significant long-run driver of disposable income, implying structural economic transmission; GDP and government transfers are also significant short-run stabilisers. Personal consumption expenditures provide no evidence of a statistically significant reverse-causal feedback effect on disposable income within this model. These results suggest that direct transfers are critical for rapid recovery, while structural economic transmission through GDP is required for sustained income growth over time.

Introduction

Disposable income is a central measure of household finance and one of the most significant indicators of economic well-being. Knowledge of how it can recover after economic shocks and how it can be maintained across long-run structural cycles can guide policymaking. According to the original theory of Keynes (1936), national output translates into household income and therefore into consumption. This macroeconomic relationship between income and consumption was given a new empirical interpretation by Friend (1946), who discussed the endogeneity of income and consumption.

Nonetheless, household income is not primarily determined by free-market dynamics alone. Government action stands as a rapid stabiliser, able to instantaneously change the allocation of income and household liquidity during crisis periods (Papadimitriou, 2006). Discretionary fiscal methods and automatic stabilisers, such as federal transfers, insulate households against demand shocks in a real-time manner (Blinder & Zandi, 2015; Taylor, 2000). Indeed, aggregate economic conditions define the broader context that households face. As shown by Okun's theory (1962), widespread economic expansion affects labour market outcomes, which in turn determine wage returns and disposable income. Current empirical evidence also supports the view that income variations affect household balance sheets and can critically influence aggregate demand (Mian et al., 2013; Romer & Romer, 2010). Despite the breadth of this literature, the joint dynamic effects of these channels on U.S. disposable income over an extended postwar horizon remain underexplored within a unified cointegrating framework—a gap this study addresses.

Guided by this model, the paper focuses on the following research question: How do macroeconomic aggregates, government transfers, and consumption dynamically affect disposable income in the U.S. between 1951 and 2025?

In alignment with the ARDL framework, we test the following hypotheses:

- H_1 : Macroeconomic conditions, represented by GDP and unemployment, share a significant long-run relationship with disposable income, with GDP expected to exert a positive effect and unemployment expected to exert a negative effect.

- H₂: Government transfers exert a significant, positive short-run impact on disposable income, acting as immediate policy stabilisers.
- H₃: PCE does not exhibit a statistically significant feedback effect on disposable income in the long run.

Methodology

Data and Variable Transformation

The time series quarterly dataset covers the period from 1951Q4 to 2025Q1, thus $n = 297$. All five variables are extracted from the Federal Reserve Economic Data database, from the Federal Reserve Bank of St. Louis: disposable personal income (DPI), gross domestic product (GDP), personal consumption expenditure (PCE), civilian unemployment rate (UNRATE), and government current transfer payments (B087RC1Q027SBEA). The series characteristics of the raw data variables are reported in Figure 2 (Appendix).

The dependent variable for this particular study is DI, which represents after-tax income that households may choose to spend as they prefer. It is the most vital measure reflecting household stability over the research period (Keynes, 1936; Friend, 1946). Four explanatory variables are included, each selected on the basis of established economic theory:

- GDP: captures the structural output-to-income transmission channel. The Keynesian framework (Keynes, 1936) establishes that aggregate output flows through labour markets into household income, implying a positive long-run relationship consistent with H₁.
- UNEMP: captures labour market conditions in the spirit of Okun's Law (1962). Higher unemployment compresses wage returns and reduces household income, suggesting a negative relationship with DI and forming the second component of H₁.
- PCE: included to test whether household spending exerts a reverse-causal feedback effect on income. Friend (1946) and Mian et al. (2013) raise the possibility of this endogenous relationship; however, the Keynesian consumption function suggests the direction runs from income to consumption rather than the reverse, motivating H₃.
- TRANSFER: captures the fiscal stabiliser mechanism. Blinder & Zandi (2015) and Taylor (2000) establish that direct government transfers inject liquidity into household

budgets with relatively little delay, particularly during economic contractions, motivating the short-run focus of H2.

Considering that the four variables DI, GDP, PCE, and TRANSFER are positive dollar financial flows and constitute a considerable increase over the 74-year time period, we decided to apply the natural logarithm transformation to them. This allows for stabilizing variance, reducing scale differences, and elastic interpretation of coefficients in the results as seen in Figure 1 (Appendix), which depicts more symmetrical distributions. The unemployment rate is retained in percentage points, as it is also a rate; a logarithmic transformation did not apply to this variable, nor does it make any economic sense, and the unemployment rate coefficient is interpreted semi-elastically throughout the paper.

Model Specification

This paper employs an ARDL specification to model the dynamic relationship between US disposable income and its macroeconomic fundamentals. We employ an ARDL specification instead of OLS static estimation because we need a dynamic adjustment process for macroeconomic variables to move over time and over a VAR because the variables have mixed orders of integration. Since the results of the unit root tests show that there are variables that are I(0) and I(1), the Pesaran et al. (2001) specification is the most appropriate.

The general ARDL(p, q₁, q₂, q₃, q₄) model is specified as:

$$\begin{aligned} \log(DI_t) = & \alpha_0 + \sum_{i=1}^p \Delta \phi_i \log(DI_{t-i}) + \sum_{j=0}^{q_1} \Delta \beta_{1,j} \log(GDP_{t-j}) + \sum_{j=0}^{q_2} \beta_{2,j} (UNEMP_{t-j}) \\ & + \sum_{j=0}^{q_3} \Delta \beta_{3,j} \log(PCE_{t-j}) + \sum_{j=0}^{q_4} \Delta \beta_{4,j} \log(TRANSFER_{t-j}) + u_t \end{aligned}$$

Where all variables are log variables from the above section, and p and q₁-q₄ are lags from AIC selection, and u_t refers to a white noise error term.

For cointegration and long-run estimation, it can be reparameterized as an unrestricted error correction model:

$$\Delta DI_t = \alpha_0 + \rho DI_{t-1} + \sum_{k=1}^4 \beta_k X_{k,t-1} + (\text{short-run terms}) + \varepsilon_t$$

Where ρ is the speed-of-adjustment coefficient and β_1 - β_4 are the long-run level coefficients on GDP, UNEMP, PCE, and TRANSFER, respectively. The short-run multipliers

are derived from the first-difference terms of the UECM, while the long-run multipliers are calculated using the ratios of the level terms. The optimal lag orders from the AIC method are tabulated in the next section.

Unit Root Testing Procedure

Before estimating the ARDL model, we first checked whether the variables were stationary. This step was necessary because the ARDL bounds test can be used when the variables are $I(0)$, $I(1)$, or a mixture of both, but it should not be applied if any variable is $I(2)$. Therefore, we had to identify the integration order of each variable before moving to the bounds test.

We first looked at the time-series plots to get a visual idea of whether the variables had clear trends or moved around a stable range. However, the plots were only used as an initial check, so we then applied formal unit root tests. The Augmented Dickey-Fuller test was used as the main test, where the null hypothesis is that the series has a unit root and is non-stationary. If the p-value is below 0.05, we reject the null and treat the series as stationary. The Phillips-Perron test was also used as a robustness check.

ARDL Bounds Testing Procedure

Prior to embarking on the short-run and long-run multipliers test, it is necessary to determine if there exists a long-run relationship among the variables. This study employs the ARDL bounds testing approach of Pesaran, Shin, and Smith (2001), which tests cointegration by checking whether or not the level terms for all the variables in the UECM are indeed statistically significant. This process, in contrast to the traditional cointegration techniques embodied in the Engle-Granger two-step procedure or the Johansen trace test, is not affected by the presence of $I(0)$ and $I(1)$ variables. We carried out the unit root tests, and as a result, we found that the variables were mostly $I(0)$ or $I(1)$. Therefore, the most important condition that is normally expected for the ARDL bounds testing approach, the absence of the order two $I(2)$ variable, was met.

The lag order of the UECM is selected using the AIC criterion with a maximum of 4 lags per variable. The dynamic specification of the model is appropriate, as the data are quarterly macroeconomic data with high persistence, and the adjustment mechanisms (for example, the adjustment of household income to GDP growth) span several quarters. The bounds F-test conducted under Case 3 of the Pesaran et al. (2001) framework has an intercept, but there is no deterministic trend. This is suitable for our ARDL model, as there is an intercept but not a deterministic trend; there is only a stochastic trend and not a deterministic one, as depicted in the time-series graphs (Figure 2, Appendix).

The hypotheses for the bounds F-test are the following:

$H_0: \rho = \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$ (no cointegration relationship in the long run)

H_1 : at least one level coefficient is non-zero (there is a cointegration relationship in the long run)

The testing procedure criteria are as follows: if the computed F-statistic exceeds the 5% I(1) upper critical bound, then reject H_0 and accept cointegration; if the computed F-statistic is less than the 5% I(0) lower critical bound, then do not reject H_0 ; an F-statistic between the two bounds yields an inconclusive result.

ARDL Estimation and Lag Selection

The optimal lag structure is determined using the AIC, with a maximum of four lags per variable. The maximum lag length of four is set in accordance with the quarterly frequency of the data, allowing up to one full calendar year of lagged adjustment — a standard convention for quarterly macroeconomic time series. Lag selection is conducted over the unrestricted ARDL levels specification prior to reparameterisation as the UECM.

AIC is preferred over the Bayesian Information Criterion (BIC) because BIC's heavier penalty on additional parameters risks under-fitting the lag structure in the context of quarterly macroeconomic data with highly persistent dynamics. Where income, output, and fiscal variables adjust gradually across multiple quarters, an overly parsimonious model may fail to capture economically meaningful delayed adjustments. AIC therefore prioritises fit sufficiency

over model brevity, ensuring that the distributed-lag dynamics of variables such as TRANSFER and GDP are fully accommodated in the selected specification.

The AIC selected an ARDL(3,3,0,3,4) model, denoting three lags on DI, three lags on GDP, zero lags on UNEMP, three lags on PCE, and four lags on TRANSFER respectively.

The zero-lag assignment on UNEMP reflects both a theoretical and statistical rationale. Theoretically, Okun's Law (1962) positions unemployment as an indirect channel operating through GDP rather than as an independent driver of DI; once aggregate output is controlled for in the model, the marginal information contained in lagged unemployment values is largely redundant, as its effect on DI is already captured by the GDP coefficient. Statistically, the AIC confirms this by assigning zero additional lags, indicating that lagged values of UNEMP offer no meaningful improvement in model fit. This implies that unemployment exerts a contemporaneous rather than a distributed-lag effect on disposable income.

The four-lag assignment on TRANSFER reflects the well-documented fiscal transmission delay inherent in government disbursement programs. Whilst transfer payments are rapid relative to other fiscal instruments, they remain subject to administrative pipelines, eligibility determination, and disbursement scheduling that distribute their income effects across multiple quarters. A lag length of four quarters — equivalent to one full calendar year — is therefore theoretically consistent with the literature on automatic stabilisers (Blinder & Zandi, 2015; Taylor, 2000), which establishes that the income effects of transfer programs are not fully absorbed within a single quarter.

Following lag selection, the ARDL(3,3,0,3,4) model is reparameterised as a UECM, from which short-run and long-run multipliers are derived. The long-run multipliers are extracted from the level terms of the UECM using the ratio method, whilst the short-run multipliers are obtained from the first-differenced terms capturing quarter-on-quarter adjustments. Given that the Breusch-Godfrey test subsequently confirmed the presence of serial correlation in the residuals, HAC (Newey-West) standard errors are applied throughout the estimation to ensure that inference on both the short-run and long-run coefficients remains valid under autocorrelated errors.

Serial Correlation/Autocorrelation Tests

To check if our model contains autocorrelation, we first convert our ARDL model to a linear model. We would create a correlogram which displays the ACF and PACF as shown in Figure 3 (in appendix).

For $n = 297$, the approximate 95% significance bounds are $\pm \frac{1.96}{\sqrt{297}}$. Lags that are within the bounds are white noise, and lags that are over the bounds are significant for autocorrelation.

The PACF shows a significant spike at Lag 3 so we would perform a Breusch-Pagan test on order 3. Furthermore, since our time-series is in quarterly data it would also be appropriate to perform a Breusch-Pagan test on order 4.

We perform Breusch-Pagan test on order 3 and order 4;

Hypotheses:

H_0 : *Errors are NOT serially correlated*

H_1 : *Errors are Serially Correlated*

BG(3); P-value ≈ 0.0050

BG(4); P-value ≈ 0.0117 (Chi-square), P-value ≈ 0.0155 (F stat method)

Since BG(3) and BG(4) p-values < 0.05 , we reject the null hypothesis and conclude that there is statistically significant evidence of serial correlation in our model.

Since serial correlation exists, our standard errors may be incorrect which invalidates our t-tests and F-tests. We have to correct the SEs by using HAC (Newey West) Standard Errors.

Looking at Output 1 and Output 2 (in appendix). All variables still have the same p-value significance/insignificance. The coefficients for all variables are still very similar, i.e. no major differences between the OLS estimates and the OLS with HAC estimates. The interpretation of every variable will still be the same. The long-run multipliers were similarly re-estimated using HAC standard errors via `vcovHAC(to_lm(best_model))`, consistent with the levels parameterisation from which the multipliers are derived. GDP's long-run significance is retained ($p = 0.0043$), while all other variables remain insignificant, confirming that the basis-matched HAC correction does not alter the substantive conclusions.

However, when looking at the standard errors (Output 3 in appendix), Standard error increases for variables such as Disposable Income(DI) at lag 1 to lag 3, Gross

Domestic Product (GDP) at lag 0, lag 2, lag 3, Personal Consumption Expenditure (PCE) at lag 0, lag 2, lag 4, and Federal Government Transfers (Transfer) at all lags. OLS SEs were understated for these variables. For the rest of the variables, OLS SEs were overstated, or nothing changed. The variables and their lagged parts that have significant changes in their SEs shows that those variables are sensitive to autocorrelation indicating that these variables rely on their previous values.

Heteroskedasticity Tests

From Figure 4 (in appendix), there isn't an obvious fanning structure when looking at the residuals plot, but there does look like there exists a few outliers which we cannot remove as it will ruin our time series structure. However, we will still be conducting a Breusch-Pagan test to confirm whether heteroskedasticity exists in our model.

Hypotheses:

$$\begin{aligned}H_0: & \text{Homoskedasticity} \\H_1: & \text{Heteroskedasticity}\end{aligned}$$

Breusch-Pagan p-value: 0.0739. Therefore, on a 0.05 significance level we retain the null hypothesis since $0.0739 > 0.05$. We conclude that there is insignificant statistical evidence that heteroskedasticity exists in our model.

Before conducting our heteroskedasticity tests, we had transformed our model to a log-log model specification which had resulted in variance being more stable which reduces the severity of heteroskedasticity in our ARDL model. This is backed up by our Breusch-Pagan test with a P-value of 0.0739 which indicates that there is insignificant statistical evidence of heteroskedasticity present in our model.

Diagnostic and Stability Check Methods

After estimating the ARDL model, we also checked whether the model was reliable for interpretation and inference. This was important because even if the model produces significant coefficients, the results may still be misleading if the residuals show serial correlation, heteroskedasticity, non-normality, or instability.

We used the Breusch-Godfrey test to check whether the residuals were serially correlated. If serial correlation is present, the usual standard errors may be unreliable, which can affect the t-tests and p-values. For this reason, HAC/Newey-West standard errors were applied when serial correlation was detected. We also used the Breusch-Pagan test to check for heteroskedasticity, where the null hypothesis is homoskedasticity. In addition, the Ramsey RESET test was used to check whether there was strong evidence of functional form misspecification, and the Jarque-Bera test was used to assess residual normality. Finally, CUSUM and MOSUM stability tests were used to check whether the model appeared stable over the sample period.

Results and Discussion

Unit Root Test Results

Before running the formal unit root tests, we first looked at the time-series plots of the variables. The plots show that DI, GDP, PCE and federal government transfer generally move upward over time, while unemployment fluctuates within a more stable range. This suggested that some variables may not be stationary at level, but we could not rely on the plots alone. Therefore, we used the ADF test to formally check the stationarity of each variable.

The ADF results are reported in Table 1. For the unemployment rate, the p-value at level is 0.0228, which is below 0.05. Therefore, we reject the null hypothesis of a unit root and classify UNEMP as $I(0)$. For DI, GDP, PCE, and federal government transfer, the p-values at level are greater than 0.05, so we fail to reject the unit root null at level. After first differencing, all four variables have p-values below 0.01, so they are treated as stationary after the first difference and classified as $I(1)$.

The PP test was also used as a robustness check and supported the same classification. Overall, the variables are a mixture of $I(0)$ and $I(1)$, with no variable found to be $I(2)$. This is important because the ARDL bounds testing approach can be used with $I(0)$, $I(1)$, or mixed $I(0)/I(1)$ variables, but not with $I(2)$ variables. Therefore, the unit root results support moving to the ARDL bounds testing stage.

Bounds Test Results

As alluded to in the previous section, results of the unit root test revealed a combination of $I(0)$ and $I(1)$ variables with no $I(2)$ series detected; thus, the ARDL bounds F-test was performed to investigate the existence of a long-run cointegrating relationship among the five variables. Lag selection based on the AIC criterion selected an ARDL(3,3,0,3,4) model. The absence of lag for UNEMP indicates that unemployment impacts disposable income contemporaneously and does not need to be lagged. The dynamics of TRANSFER (4) reflect the long fiscal impulse response of transfer programs, which can take a couple of quarters for households to feel their effect. This dynamic nature is appropriate given the use of quarterly data and the apparent macroeconomic dynamics during the sample period.

Please refer to Table 3. Bounds F-test Result in the Appendix.

As the calculated value of the F-statistic (6.40) is significantly larger than the upper critical value (5% - 4.85), the null hypothesis is rejected. There is statistical evidence for an actual long-run cointegration relationship. This means that GDP, transfers, unemployment, PCE, and disposable income are not only showing a short-run relationship but are also converging to the same long-run relationship with the passage of time as well.

From the economic point of view, it suggests that the variables are cointegrated, implying that there exists a stable equilibrium path for the long run. While there may be deterioration, rise, or even expansion of the series overrides during the short run due to income shock during recession and fiscal expansion, desynchronized by the error-correction term, the system is always dragged back along the equilibrium long-run path. This finding also confirms the economic principle embedded in the very rationale for undertaking this study in the first place. Precisely, the transmission of income to output and fiscal stabilization as conceived by Keynesian economics is not just temporal but a structural long-run emerging pattern of the U.S. economy in the post-war period. The finding gives some evidence in support of H_1 that macroeconomic variables and government transfers have, in general, a long-run association with the households' disposable income as such. The long-term significance of the individual effects of unemployment and personal consumption expenditure will be evaluated in the next section through ARDL coefficient estimations.

ARDL Short-Run and Long-Run Results

Long-Run Estimates

The long-run multipliers were extracted from the level terms of the UECM using the `multipliers()` function. Given that serial correlation was detected and confirmed by the Breusch-Godfrey test, HAC (Newey-West) standard errors were applied throughout to ensure valid inference on the long-run coefficients.

Under HAC (Newey-West) robust standard errors, GDP remains the sole statistically significant long-run determinant of disposable income ($\beta = 0.680$, $p = 0.0043$) consistent with Keynesian income theory whereby aggregate output transmits into household income through labour market channels including wages, employment levels, and profit distributions (Bellino, 2021). A 1% rise in GDP translates to a 0.68% increase in DI over the long run. The multiplier being less than unity suggests that a portion of output gains is absorbed by corporate retained earnings, taxes, and savings rather than passed directly to households (Mian et al., 2013).

Government transfers exhibit a positive but statistically insignificant long-run effect ($\beta = 0.061$, $p = 0.068$). As transfer payments are counter-cyclical by design and function as automatic stabilisers rather than permanent income sources, their effects do not accumulate into a sustained long-run income change — consistent with the Permanent Income Hypothesis, which holds that households respond more strongly to persistent rather than temporary income shifts (Krueger et al., 2023). A competing explanation is that the long-run insignificance may partly reflect aggregation bias, where pooling transfer payments across 74 years of heterogeneous fiscal regimes understates the structural significance of transfers during specific sub-periods such as the 2008 Global Financial Crisis or the COVID-19 pandemic.

PCE ($\beta = 0.254$, $p = 0.230$) and UNEMP ($\beta = 0.002$, $p = 0.487$) are both statistically insignificant in the long run. The insignificance of UNEMP is grounded in Okun's Law (1962), which positions unemployment as an indirect channel through GDP rather than an independent income driver — once GDP is controlled for, UNEMP becomes redundant. The insignificance of PCE is consistent with the Keynesian consumption function, where causality runs from DI to PCE rather than the reverse (Keynes, 1936). A potential endogeneity concern arises given

that PCE and DI may be simultaneously determined within a single-equation framework; PCE is therefore better understood as a control variable rather than a causal regressor.

Short-Run Dynamics

GDP exhibits the most pronounced short-run influence, with a contemporaneous effect of 0.370 ($p < 0.0001$), indicating that a 1% increase in GDP raises DI by 0.37% within the same quarter. This effect persists across two additional quarters — $\Delta(L(\text{GDP},1)) = 0.211$ ($p = 0.006$) and $\Delta(L(\text{GDP},2)) = 0.230$ ($p = 0.002$) — confirming that GDP shocks transmit gradually over approximately three quarters before converging toward the long-run multiplier of 0.68%, reflecting the time required for wage renegotiations, hiring decisions, and income redistribution to fully materialise (Sanso-Navarro & Vera-Cabello, 2020).

Government transfers demonstrate immediate short-run transmission, with a contemporaneous effect of 0.160 ($p < 0.0001$). Additional significant lagged effects at quarter one (0.044, $p = 0.001$) and quarter three (0.030, $p = 0.0002$) suggest the administrative disbursement pipeline extends transfer effects across four quarters, consistent with H_2 . UNEMP generates no statistically significant short-run effects, reinforcing the Okun's Law interpretation. Only one lagged PCE term achieves significance — $\Delta(L(\text{PCE},2)) = -0.162$ ($p = 0.033$) — though this should be interpreted cautiously given the broader insignificance of PCE across all other lag terms.

Error Correction Mechanism

The ECT is -0.157 ($p < 0.0001$), confirming a valid error correction mechanism whereby 15.7% of any disequilibrium is corrected each quarter. The mean adjustment lag of 6.4 quarters reflects structural rigidities in labour markets and government transfer systems that prevent immediate equilibrium restoration following income shocks (Sanso-Navarro & Vera-Cabello, 2020).

Limitations

The single-equation ARDL framework cannot resolve structural causation or fully address endogeneity, particularly for PCE. Residual non-normality was confirmed by the

Jarque-Bera test ($p < 2.2e-16$), though the large sample size mitigates this for asymptotic inference. The model also does not account for structural breaks including the 2008 Global Financial Crisis and COVID-19 pandemic. Future research could consider an NARDL specification or break dummies to capture asymmetric responses and regime changes across the sample period.

Diagnostic and Stability Check Results

After estimating the ARDL model, we checked whether the model residuals and parameters were reliable enough for interpretation. The results are summarised in Table 2. These tests were used because a model can still give misleading inferences if there are problems such as serial correlation, heteroskedasticity, non-normal residuals, or instability.

The Breusch-Godfrey test shows evidence of higher-order serial correlation, since the higher-order (order 3 & order 4) p-values are below 0.05. This means that the residuals are not fully independent over time. Because serial correlation can make the usual standard errors unreliable, HAC/Newey-West standard errors were applied. This does not remove the serial correlation itself, but it helps make the inference more reliable by adjusting the standard errors.

These autocorrelated errors are consistent with economic theory. Disposable income gradually evolves over time, as according to Sanso-Navarro & Vera-Cabello (2020), disposable income changes its trend based on labour market conditions, changes in minimum wage and pensions, and cash transfers.

For heteroskedasticity, the Breusch-Pagan test gives a p-value of 0.0739, which is greater than 0.05. Therefore, we fail to reject the null hypothesis of homoskedasticity, and there is no strong statistical evidence of heteroskedasticity in the model. The Ramsey RESET test also gives a p-value of 0.1074, which is above 0.05. This suggests that there is no strong evidence of functional form misspecification.

By logging the variables such as Disposable income (DI), Gross Domestic Product (GDP), Personal Consumption Expenditure (PCE), and Government Transfer (Transfer), the variation in these variables becomes more stable as we have converted the variables and the model from dollar-based (US\$) to percentage-based. With the variance being more stable on a

log-log model, we were already able to reduce the effects of heteroskedasticity which was proven by our Breusch-Pagan test.

However, the Jarque-Bera test rejects residual normality, with a p-value below $2.2e-16$. This means that the residuals are not normally distributed, so this is acknowledged as a limitation. Since the sample size is relatively large, this issue is less serious for asymptotic inference, but it should still be considered when interpreting the results.

Conclusion

The current study investigated the dynamic drivers of U.S. disposable income from 1951Q4 to 2025Q1, generating findings of considerable significance for macroeconomic theory and policy. The ARDL bounds test indicated a clear longstanding relationship ($F = 6.40$, $> 5\%$ upper bound of 4.85), and CUSUM and MOSUM parameter stability tests demonstrated overall stability, with results estimated using HAC/Newey-West standard errors due to serial correlation.

H_1 is partially supported. Under HAC-robust inference, GDP remains the only statistically significant long-run predictor of disposable income ($\beta = 0.680$, $p = 0.0043$). A 1% increase in GDP translates to a 0.68% increase in disposable income, consistent with Keynesian income theory and validating the long-run structural transmission channel posited by Keynes (1936). In the end, GDP (as an illustration of growth) does make its entry into household wealth indirectly through wages and added occupations. In contrast, unemployment was not detected as a significant predictor of the long-run model at the 5% threshold, which signifies that structural change is more concentrated on the output side than on the labour market side across the full sample period.

H_2 is fully supported. In the short run, effects work through two channels. GDP lifts disposable income today by 0.37% through instantaneous market clearing, while the federal government transfer acts as an additional stabiliser through a direct 0.16% effect today itself. This confirms federal transfers' role as a fiscal stabiliser as hypothesized, consistent with Blinder & Zandi (2015) and Taylor (2000). The autocorrelated shocks to the disposable income that persist in time, matching the empirical income persistence trend documented by Sanso-

Navarro & Vera-Cabello (2020), help to explain why short-run stabilising forces are even needed in the backdrop of long-term structural effects.

H₃ is supported: PCE produced statistically insignificant results in both periods. This could be interpreted in the context of a claim made by Friend in 1946, who stated that it is disposable income that causes consumption and not the other way around. However, this needs to be treated with caution, as one cannot completely segregate income and consumption in an ARDL single-equation model; the insignificance of PCE might be due to endogeneity rather than the actual absence of feedback.

These policy implications are important because of the United Nations' Sustainable Development Goals. The fact that GDP has continued to persist indicates that SDG 8 (Decent Work and Economic Growth), which states that sustained and broad-based economic growth is the most effective way to increase household income over the structure cycle, is more valid. Nevertheless, GDP cannot be generated quickly during a sudden crisis (for instance, recession or pandemic), and this is where the immediate impact of government transfers becomes critical. This highlights an additional important channel for SDG 10 (Reduced Inequalities) and SDG 1 (No Poverty). Direct fiscal transfers are, therefore, the main policy tools to inject liquidity and safety nets to households when private markets fail.

A downside of the study is that the ARDL approach identifies dynamic relationships conforming to economic theory; however, the model does not necessarily imply structural causation. The model also required HAC correction for serial correlation, and the Jarque-Bera test indicates the presence of non-normality of errors; thus, the inference should be drawn carefully. The study by no means accounts for any structural break, including the 2008 Global Financial Crisis or the Covid-19 pandemic, that could have structurally changed the income-generating process. Future works can consider investigating a nonlinear ARDL for testing if the positive and negative GDP shocks have a differential effect on disposable income or could include break dummies to account for known structural breaks that provoke regime changes. Understanding these economic non-linearities can assist in fabricating a detailed fiscal policy response.

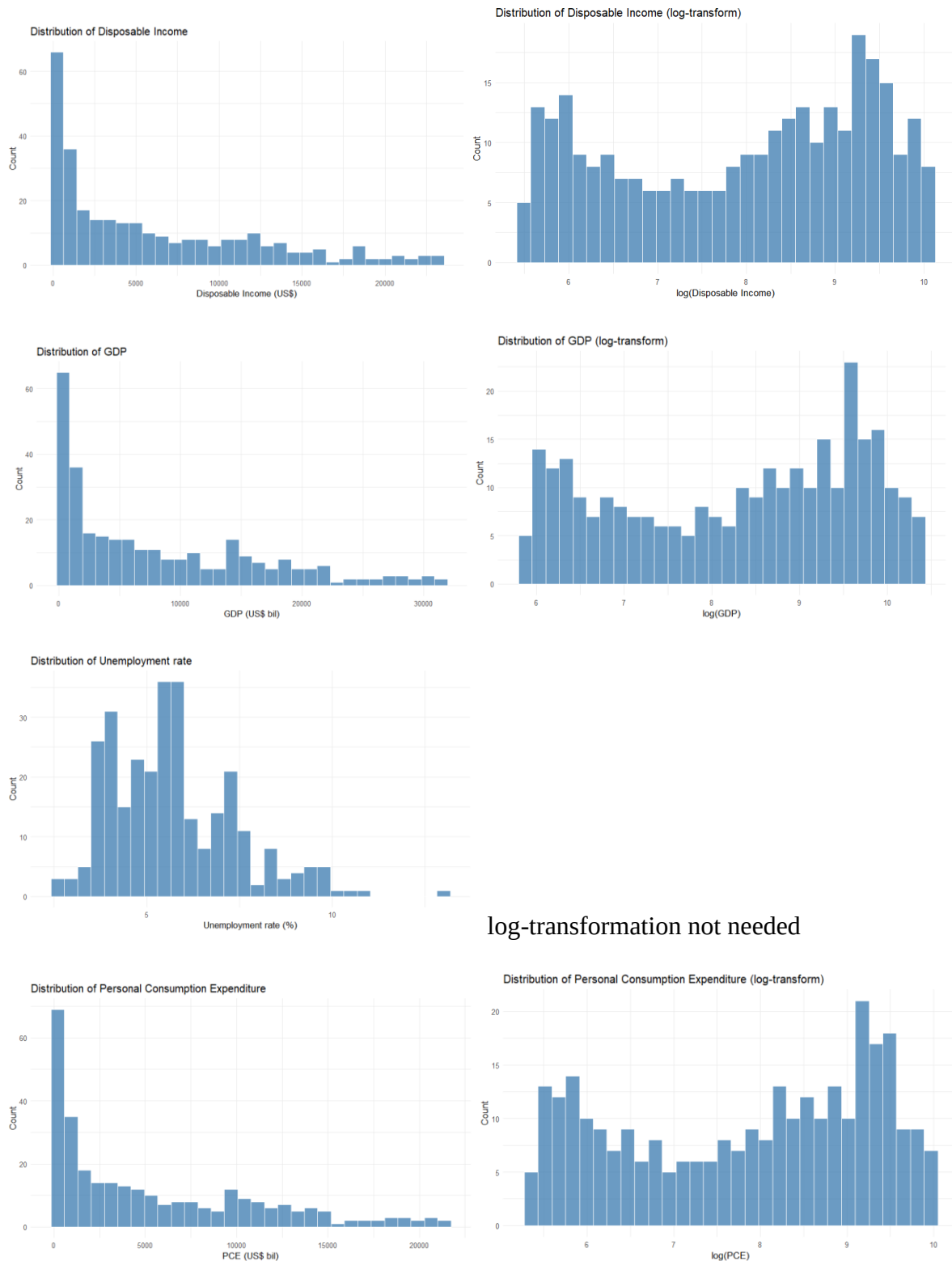
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Appendix

Figure 1: Histograms of chosen variables, before and after log-transformation



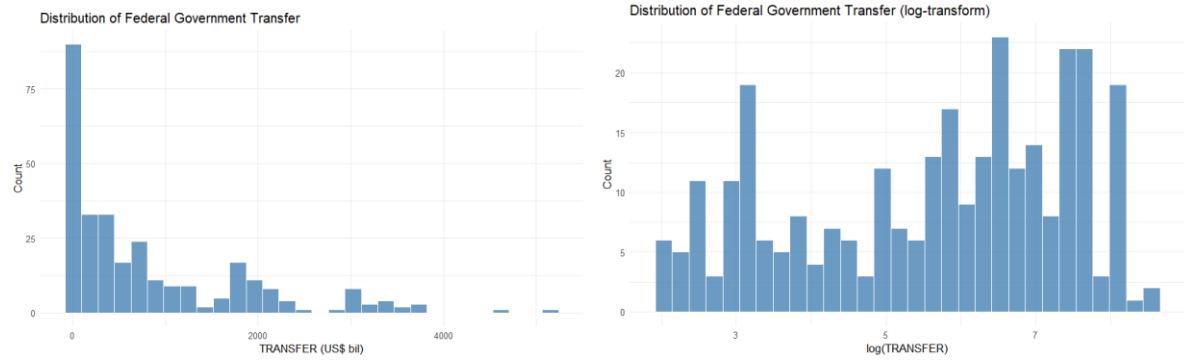


Figure 2. Time-series plots of selected variables

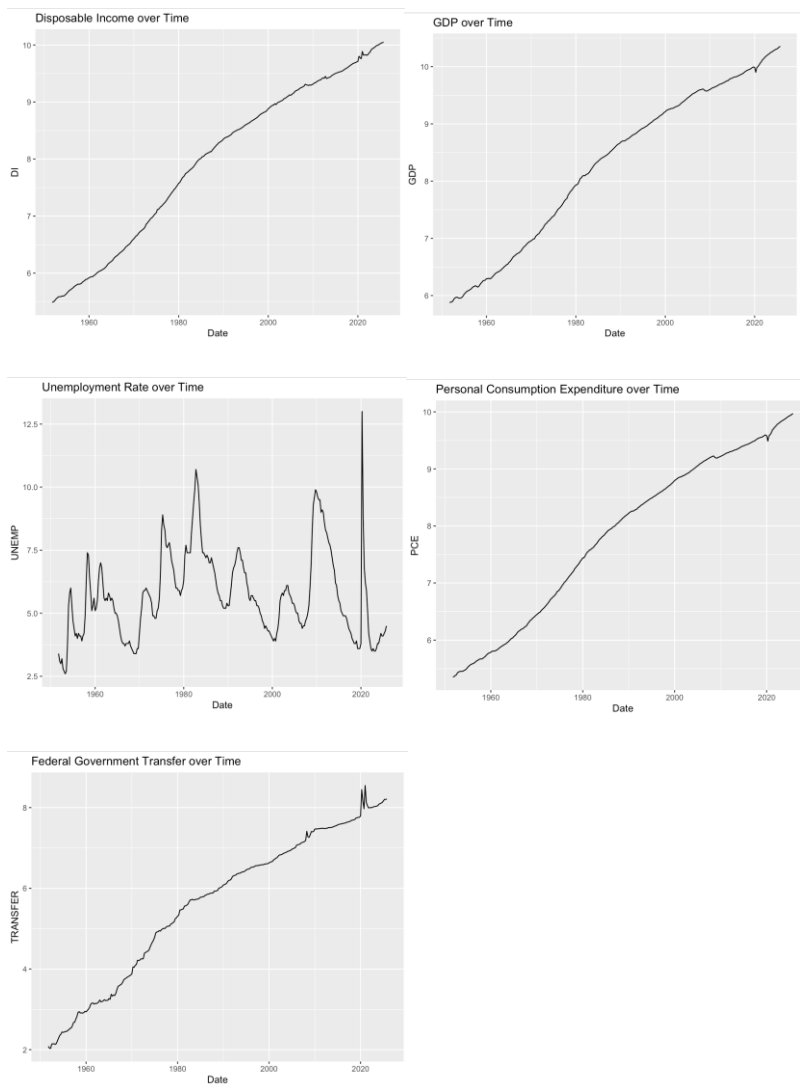


Figure 3: Visual diagnostics using correlogram

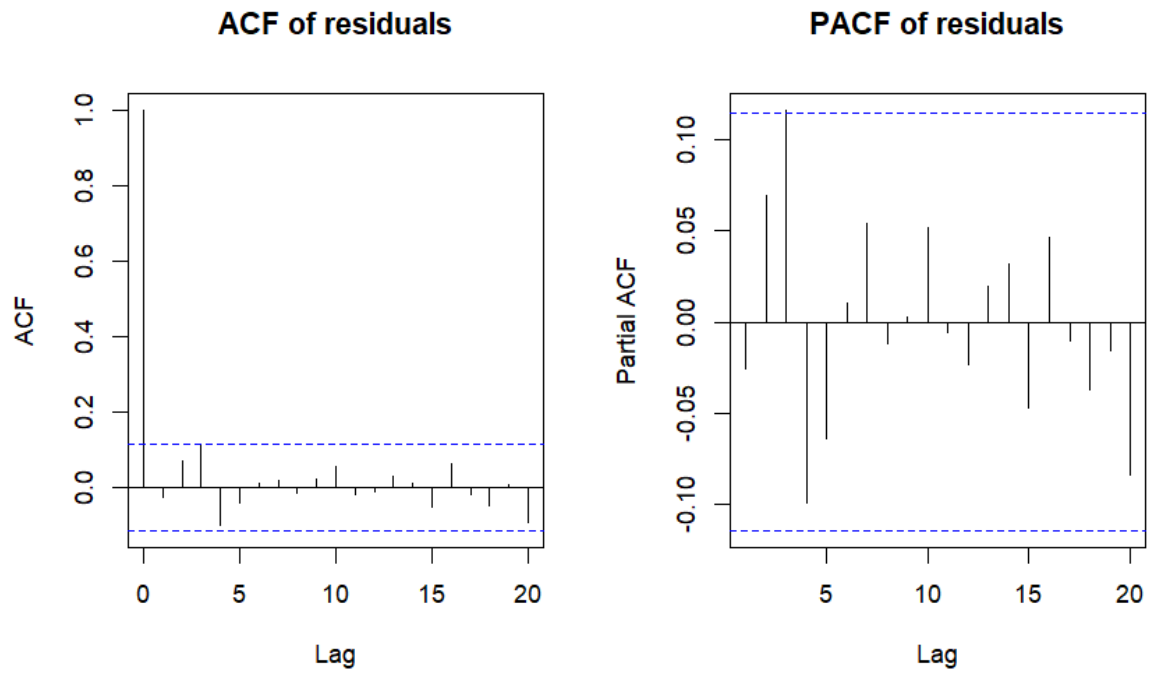


Figure 4: Residuals plot

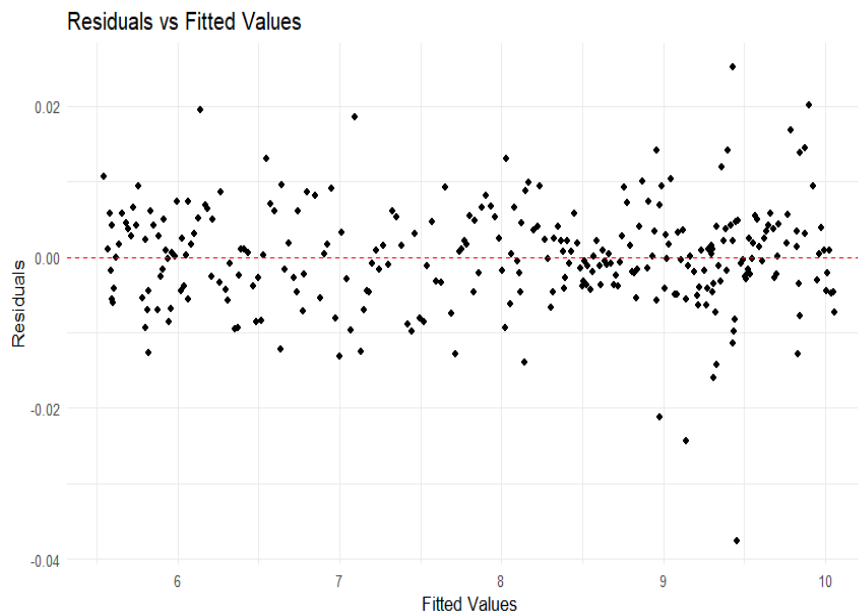
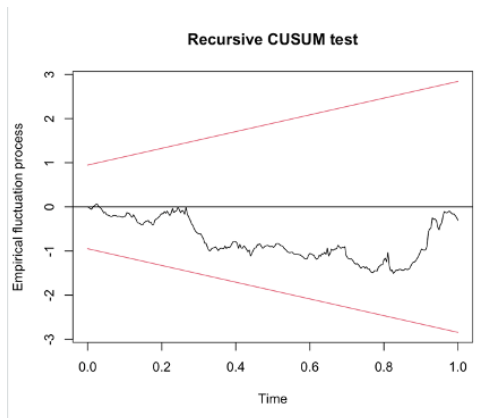


Figure 5. CUSUM Stability Plot



Output 1: OLS estimates and Standard Errors

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-3.098e-03	4.890e-02	-0.063	0.949531	
L.DI.1	7.160e-01	5.776e-02	12.397	< 2e-16	***
L.DI.2	2.622e-01	7.032e-02	3.729	0.000234	***
L.DI.3	-1.352e-01	5.579e-02	-2.424	0.015995	*
GDP	3.700e-01	7.354e-02	5.031	8.84e-07	***
L.GDP.1	-5.256e-02	9.541e-02	-0.551	0.582106	
L.GDP.2	1.957e-02	9.321e-02	0.210	0.833844	
L.GDP.3	-2.302e-01	7.400e-02	-3.110	0.002066	**
UNEMP	3.055e-04	4.252e-04	0.718	0.473073	
PCE	-3.934e-05	7.904e-02	0.000	0.999603	
L.PCE.1	1.454e-02	9.850e-02	0.148	0.882751	
L.PCE.2	-1.366e-01	9.855e-02	-1.386	0.166800	
L.PCE.3	1.620e-01	7.541e-02	2.149	0.032523	*
TRANSFER	1.602e-01	8.892e-03	18.019	< 2e-16	***
L.TRANSFER.1	-1.064e-01	1.417e-02	-7.512	8.20e-13	***
L.TRANSFER.2	-4.299e-02	1.467e-02	-2.931	0.003668	**
L.TRANSFER.3	2.828e-02	1.322e-02	2.138	0.033386	*
L.TRANSFER.4	-2.951e-02	7.697e-03	-3.834	0.000157	***

Output 2: OLS with Newey West Standard Errors

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-3.0981e-03	4.3754e-02	-0.0708	0.943602	
L.DI.1	7.1601e-01	9.1175e-02	7.8532	9.141e-14	***
L.DI.2	2.6222e-01	1.0098e-01	2.5969	0.009913	**
L.DI.3	-1.3524e-01	6.0773e-02	-2.2254	0.026865	*
GDP	3.6996e-01	7.5167e-02	4.9218	1.479e-06	***
L.GDP.1	-5.2565e-02	9.4781e-02	-0.5546	0.579623	
L.GDP.2	1.9571e-02	1.0239e-01	0.1911	0.848556	
L.GDP.3	-2.3016e-01	7.5485e-02	-3.0490	0.002520	**
UNEMP	3.0547e-04	4.2152e-04	0.7247	0.469251	
PCE	-3.9335e-05	1.0237e-01	-0.0004	0.999694	
L.PCE.1	1.4541e-02	9.3401e-02	0.1557	0.876395	
L.PCE.2	-1.3661e-01	1.0996e-01	-1.2424	0.215163	
L.PCE.3	1.6204e-01	7.7397e-02	2.0936	0.037215	*
TRANSFER	1.6023e-01	1.3585e-02	11.7945	< 2.2e-16	***
L.TRANSFER.1	-1.0642e-01	1.9011e-02	-5.5977	5.253e-08	***
L.TRANSFER.2	-4.2994e-02	1.8272e-02	-2.3531	0.019323	*
L.TRANSFER.3	2.8275e-02	1.4361e-02	1.9689	0.049972	*
L.TRANSFER.4	-2.9506e-02	1.0891e-02	-2.7092	0.007168	**

Output 3: Side-by-side comparison of SEs

	OLS	HAC
(Intercept)	0.0489	0.0438
L.DI.1	0.0578	0.0912
L.DI.2	0.0703	0.1010
L.DI.3	0.0558	0.0608
GDP	0.0735	0.0752
L.GDP.1	0.0954	0.0948
L.GDP.2	0.0932	0.1024
L.GDP.3	0.0740	0.0755
UNEMP	0.0004	0.0004
PCE	0.0790	0.1024
L.PCE.1	0.0985	0.0934
L.PCE.2	0.0985	0.1100
L.PCE.3	0.0754	0.0774
TRANSFER	0.0089	0.0136
L.TRANSFER.1	0.0142	0.0190
L.TRANSFER.2	0.0147	0.0183
L.TRANSFER.3	0.0132	0.0144
L.TRANSFER.4	0.0077	0.0109

Output 4: UECM Model

```
> # --- Extract UECM (Unrestricted ECM) ---
> uecm_model <- uecm(best_model)
> summary(uecm_model)
```

Time series regression with "ts" data:
Start = 5, End = 297

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
end = end)

Residuals:

	Min	1Q	Median	3Q	Max
	-0.037602	-0.003871	0.000106	0.003999	0.025212

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-3.098e-03	4.890e-02	-0.063	0.949531	
L(DI, 1)	-1.570e-01	3.085e-02	-5.090	6.66e-07	***
L(GDP, 1)	1.068e-01	3.811e-02	2.803	0.005430	**
UNEMP	3.055e-04	4.252e-04	0.718	0.473073	
L(PCE, 1)	3.993e-02	3.865e-02	1.033	0.302463	
L(TRANSFER, 1)	9.583e-03	6.183e-03	1.550	0.122314	
d(L(DI, 1))	-1.270e-01	5.806e-02	-2.187	0.029577	*
d(L(DI, 2))	1.352e-01	5.579e-02	2.424	0.015995	*
d(GDP)	3.700e-01	7.354e-02	5.031	8.84e-07	***
d(L(GDP, 1))	2.106e-01	7.535e-02	2.795	0.005558	**
d(L(GDP, 2))	2.302e-01	7.400e-02	3.110	0.002066	**
d(PCE)	-3.934e-05	7.904e-02	0.000	0.999603	
d(L(PCE, 1))	-2.543e-02	7.584e-02	-0.335	0.737663	
d(L(PCE, 2))	-1.620e-01	7.541e-02	-2.149	0.032523	*
d(TRANSFER)	1.602e-01	8.892e-03	18.019	< 2e-16	***
d(L(TRANSFER, 1))	4.423e-02	1.321e-02	3.347	0.000932	***
d(L(TRANSFER, 2))	1.231e-03	1.229e-02	0.100	0.920298	
d(L(TRANSFER, 3))	2.951e-02	7.697e-03	3.834	0.000157	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.007057 on 275 degrees of freedom
Multiple R-squared: 0.7423, Adjusted R-squared: 0.7263
F-statistic: 46.59 on 17 and 275 DF, p-value: < 2.2e-16

Output 5: RECM Model

```
> # --- Extract Restricted ECM ---
> recm_model <- recm(best_model, case = 3)
> summary(recm_model)
```

Time series regression with "zooreg" data:
Start = 5, End = 297

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
end = end)

Residuals:

	Min	1Q	Median	3Q	Max
	-0.037602	-0.003871	0.000106	0.003999	0.025212

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-3.098e-03	1.294e-03	-2.394	0.017305	*
d(L(DI, 1))	-1.270e-01	5.477e-02	-2.318	0.021152	*
d(L(DI, 2))	1.352e-01	5.234e-02	2.584	0.010274	*
d(GDP)	3.700e-01	6.957e-02	5.318	2.15e-07	***
d(L(GDP, 1))	2.106e-01	7.374e-02	2.856	0.004618	**
d(L(GDP, 2))	2.302e-01	7.236e-02	3.181	0.001635	**
d(PCE)	-3.934e-05	7.398e-02	-0.001	0.999576	
d(L(PCE, 1))	-2.543e-02	7.382e-02	-0.344	0.730742	
d(L(PCE, 2))	-1.620e-01	7.434e-02	-2.180	0.030128	*
d(TRANSFER)	1.602e-01	8.096e-03	19.791	< 2e-16	***
d(L(TRANSFER, 1))	4.423e-02	1.230e-02	3.595	0.000383	***
d(L(TRANSFER, 2))	1.231e-03	1.178e-02	0.104	0.916873	
d(L(TRANSFER, 3))	2.951e-02	7.439e-03	3.967	9.28e-05	***
ect	-1.570e-01	2.755e-02	-5.699	3.06e-08	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.007006 on 279 degrees of freedom
(0 observations deleted due to missingness)
Multiple R-squared: 0.7423, Adjusted R-squared: 0.7303
F-statistic: 61.81 on 13 and 279 DF, p-value: < 2.2e-16

Output 6: ARDL model summary

```
> # Print ARDL regression summary
> cat("--- ARDL Model Summary ---\n")
--- ARDL Model Summary ---
> print(summary(best_model))

Time series regression with "ts" data:
Start = 5, End = 297

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
             end = end)

Residuals:
      Min       1Q   Median       3Q      Max
-0.037602 -0.003871  0.000106  0.003999  0.025212

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -3.098e-03  4.890e-02  -0.063  0.949531
L(DI, 1)       7.160e-01  5.776e-02  12.397 < 2e-16 ***
L(DI, 2)       2.622e-01  7.032e-02   3.729 0.000234 ***
L(DI, 3)      -1.352e-01  5.579e-02  -2.424 0.015995 *
GDP           3.700e-01  7.354e-02   5.031 8.84e-07 ***
L(GDP, 1)     -5.256e-02  9.541e-02  -0.551 0.582106
L(GDP, 2)      1.957e-02  9.321e-02   0.210 0.833844
L(GDP, 3)     -2.302e-01  7.400e-02  -3.110 0.002066 **
UNEMP         3.055e-04  4.252e-04   0.718 0.473073
PCE          -3.934e-05  7.904e-02   0.000 0.999603
L(PCE, 1)      1.454e-02  9.850e-02   0.148 0.882751
L(PCE, 2)     -1.366e-01  9.855e-02  -1.386 0.166800
L(PCE, 3)      1.620e-01  7.541e-02   2.149 0.032523 *
TRANSFER      1.602e-01  8.892e-03  18.019 < 2e-16 ***
L(TRANSFER, 1) -1.064e-01  1.417e-02  -7.512 8.20e-13 ***
L(TRANSFER, 2) -4.299e-02  1.467e-02  -2.931 0.003668 **
L(TRANSFER, 3)  2.828e-02  1.322e-02   2.138 0.033386 *
L(TRANSFER, 4) -2.951e-02  7.697e-03  -3.834 0.000157 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.007057 on 275 degrees of freedom
Multiple R-squared: 1, Adjusted R-squared: 1
F-statistic: 6.666e+05 on 17 and 275 DF, p-value: < 2.2e-16
```

Output 6: ARDL long-run coefficients

```
> # --- Long-run coefficients ---
> cat("\n--- Long-Run (Level) Coefficients ---\n")

--- Long-Run (Level) Coefficients ---
> print(multipliers(best_model))
      Term      Estimate Std. Error    t value    Pr(>|t|)
1 (Intercept) -0.019732194 0.312648590 -0.06311301 0.949722363
2      GDP      0.680293295 0.256168551  2.65564720 0.008377591
3     UNEMP      0.001945568 0.002770727  0.70218683 0.483156552
4      PCE      0.254325020 0.223763766  1.13657821 0.256704097
5  TRANSFER      0.061032230 0.035389730  1.72457460 0.085727802
>
> cat("\n--- Long-Run Coefficients with HAC Standard Errors ---\n")

--- Long-Run Coefficients with HAC Standard Errors ---
> print(multipliers(best_model, vcov = vcovHAC(to_lm(best_model))))
      Term      Estimate Std. Error    t value    Pr(>|t|)
1 (Intercept) -0.019732194 0.279763523 -0.07053169 0.94382174
2      GDP      0.680293295 0.236395579  2.87777504 0.00431907
3     UNEMP      0.001945568 0.002798421  0.69523796 0.48749323
4      PCE      0.254325020 0.211284765  1.20370733 0.22973783
5  TRANSFER      0.061032230 0.033305674  1.83248745 0.06796005
> |
```

Table 1. ADF Unit Root Test Results

Variable	ADF p-value at level	ADF p-value at first difference	Integration order
DI	0.9900	< 0.0100	I(1)
GDP	0.9900	< 0.0100	I(1)
UNEMP	0.0228	Not required	I(0)
PCE	0.9900	< 0.0100	I(1)
TRANSFER	0.9763	< 0.0100	I(1)

Table 2. Diagnostic and Stability Checks Results

Check	Result	Decision	Implication
Breusch-Godfrey	Higher-order p-values < 0.05	Serial correlation detected	HAC/Newey-West SE applied
Breusch-Pagan	p= 0.0739	Fail to reject H_o	No strong heteroskedasticity
Ramsey RESET	p = 0.1074	Fail to reject H_o	No strong functional form misspecification
Jarque-Bera	p < 2.2e - 16	Reject normality	Non-normal residuals acknowledged
CUSUM	p = 0.4131	stable	Within critical bounds
MOSUM	p = 0.2904	stable	Additional stability support

Table 3. Bounds F-test Result

Test	Statistic	5% I(0) lower bound	5% I(1) upper bound	Decision
Bounds F-test	6.40	3.79	4.85	Reject H_o (cointegration exists)